Python - Day-9

Remove list items:-

① Remove():

→ it is an list method which can used to remove a element

Syntax:- list.remove(value)

→ not by index

→

```
a = [10, 20, 30, 40]

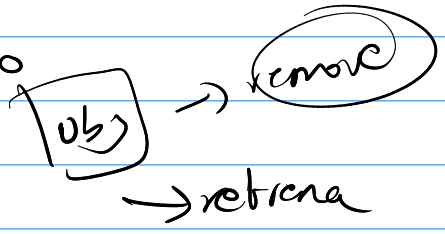
a.remove(20)

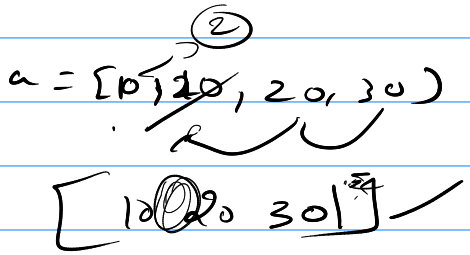
print(a)  # [10, 30, 40]
```

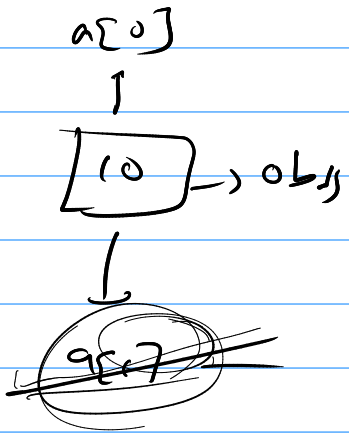
10 | 20 | 30 | 40

10 | 0 | 30 | 40

[10 | 30 | 40]







→ Time complexity

a = ["Jo", "a", "h"]

a[0] → element

a = ["J", "a", "g", "a"]

→ (1D)

→ Time complexity :- Program → how it would take to run program

a = 10, 20, 30, 40

a[0]

10

0 20 30 40 - (n)

20 20 30 40

20 30, 40

[20 30 40] (0)

O(n)

order(4)

order(n) → constant 10

, 1

pop() :-

→ it is a list a method. which can use the index and remove the element using index.

list.pop(index)

→ if not index not provided then last index element got removed

a = [10, 20, 30, 40]

a.pop()

print(a)

(1)

(0) last elem

a : [10 | 20 | 30 | 40]

(-1)

del a[-1]

→ 10 | 20 | 30 | 40

O(1)

O(1)

a = [10, 20, 30, 40]

a.pop(index)

print(a)

10 | 20 | 30 | 40

0 1 2 3

10 | 30 | 40

O(n) → Time Complexity

O(n) → Time Complexity

n → Linear Complexity

→ pop → with index → O(n)

→ pop → without index → O(1) → constant (1)

→ pop [10, 20, 30]

0 1 2

a.pop(3) → index out of range error

→ 10 | 20 | 30 | 40

0 1 2 3

10 10 20

Obj ID

10

index

it use

It not create

del keyword:

→ it is a python keyword which is used to delete obj

del list[index]

del list[start:stop]

del list

a = [10, 20, 30, 40]
0 1 2 3

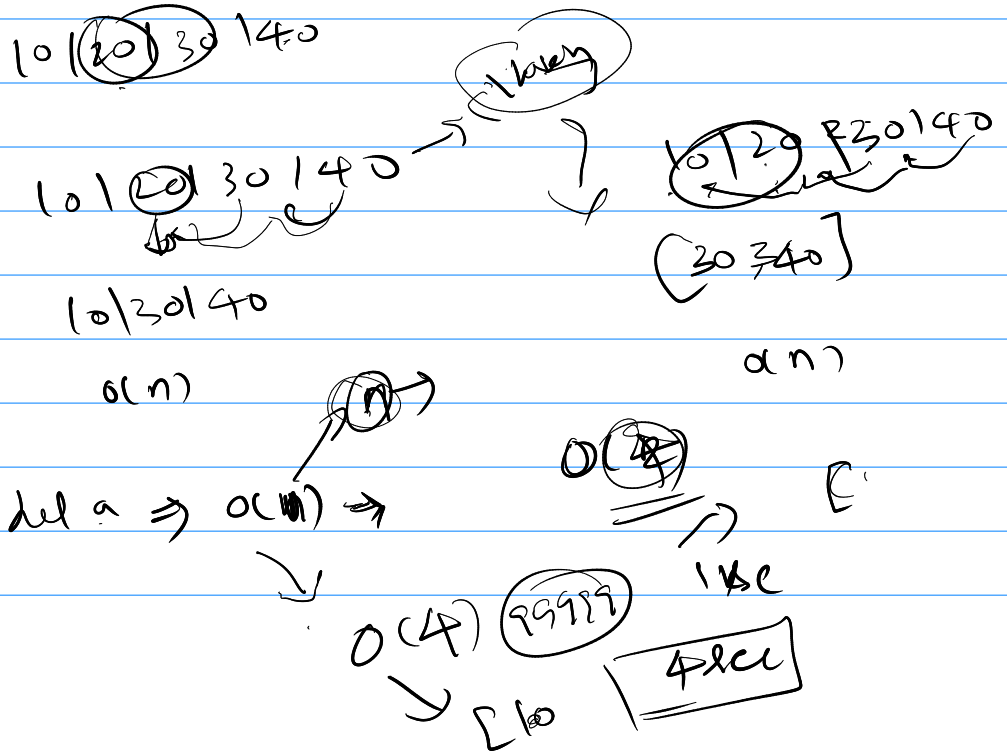
del a[0] # [20, 30, 40]

del a[0:2] # [30, 40]

del a → object deleted

↓

a not defined error



clear() :-

→ is a list method which is used to clear entre list and return a empty list

syntax

list.clear()

a = [10, 20, 30, 40] → 10001

a.clear()

print(a) → a = [ ]

→ id → size of memory

a = [10, 20, 30] = 80

a = [ ] → [ ] → 80

a = [ ]

b = [ ]

a = [10, 20, 30]

b = a

a.clear()

print(a) → true

print(b)

a = [10, 20, 30]

b = a

a = [ ]

print(a) [ ]

print(b) [10, 20, 30]

a → 10001

a = new empty list

a → 1001 → [10, 20, 30]

b →

a = [ ] → [10, 30]

1001 — print(a) [ ]

print(b) [10, 30]

List Comprehension

→ short way of writing code,
→ use existing and create a new list (iterable) → list

Syntax

newlist = [expression for i in iterable]

↳ something that we store as element in list

```
a = [1, 2, 3, 4, 5]
    1^2=1, 2^2=4, 3^2=9, 4^2=16, 5^2=25
b = [ ]
for i in a:
    b.append(i*i)
print(b)
```

* → multiplication

Dry run

a → obj → [1, 2, 3, 4, 5]
b → obj → [1, 4, 9, 16, 25]

1. [1,4]
   [1,4]
   [1,4,9]
   [1,4,9,16]

b. [].append(1*1)
   2*2
   3*3

a = [1, 2, 3, 4, 5) → iterable

```
new_list = [i*i for i in a]
print(new_list)
```

[1, 4, 9, 16, 25]

a = [1, 2, 3, 4, 5, 7]

i * i

→ [1, 4, 9, 16, 25]

Syntax

newlist = [expression for variable in iterable if condition]

→ BSpeed → BSpeed.append() → C-append

↓

Bytecode